Track-gen yield study

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Pure-Warp track-generation pipeline -- E=8192 yield study
device: cuda    regime: half_width=0.5 m, scale=10  (1 m track / ~20 m box)
pipeline: generate -> resample -> XPBD relax -> inflate  (all NVIDIA Warp)

Sweep results (end-to-end generate_tracks_warp, warmed + synced, E=8192):

lever                        yield   s/8192
-------------------------------------------
baseline 256 links/150 it    0.684    0.782

chain links (iters=150):
   128 links                 0.992    0.213
   256 links                 0.684    0.782
   384 links                 0.203    1.754
   512 links                 0.031    3.081
relax iters (256 links):
    50 iters                 0.522    0.569
   150 iters                 0.684    0.782
   300 iters                 0.766    1.105
   600 iters                 0.825    1.779
regen attempts (256 links):
    10 regen                 0.684    0.782
    20 regen                 0.684    1.162
    40 regen                 0.684    1.910
XPBD step scale sr=pr (256 links, 150 it):
  x1.0  steps                0.684    0.782
  x1.5  steps                0.341    0.785
  x2.0  steps                0.022    0.787
  x2.5  steps                0.000    0.795
  margin 0.15->0.30        0.453    0.796
  step x2.0 + margin0.30   0.000    0.789
  step x2.0, iters=50      0.023    0.565

constant_spacing (the convergence fix):
  spacing=0.30 m, N_max=384      0.999    0.557  <- yield 0.684 -> 0.999
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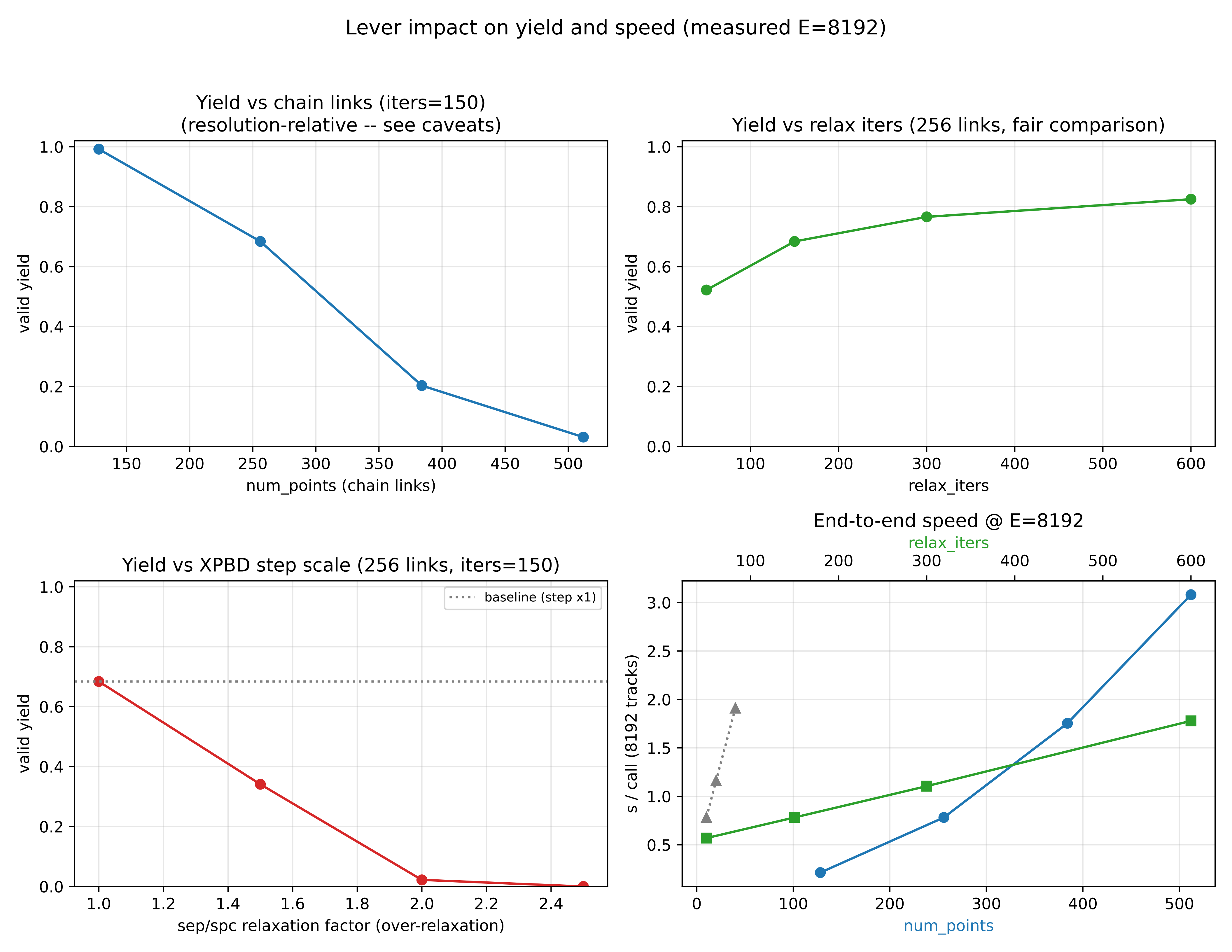
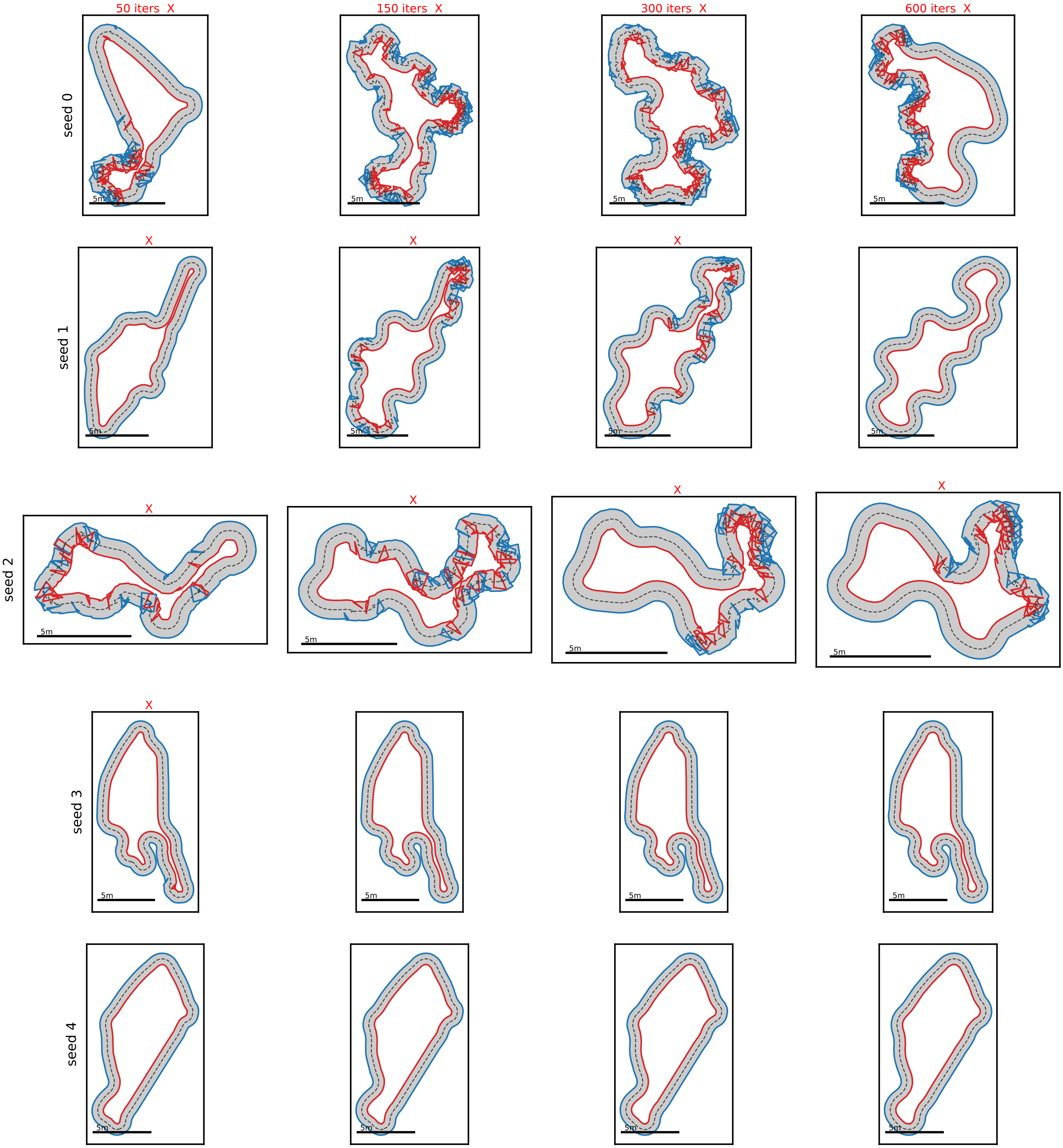
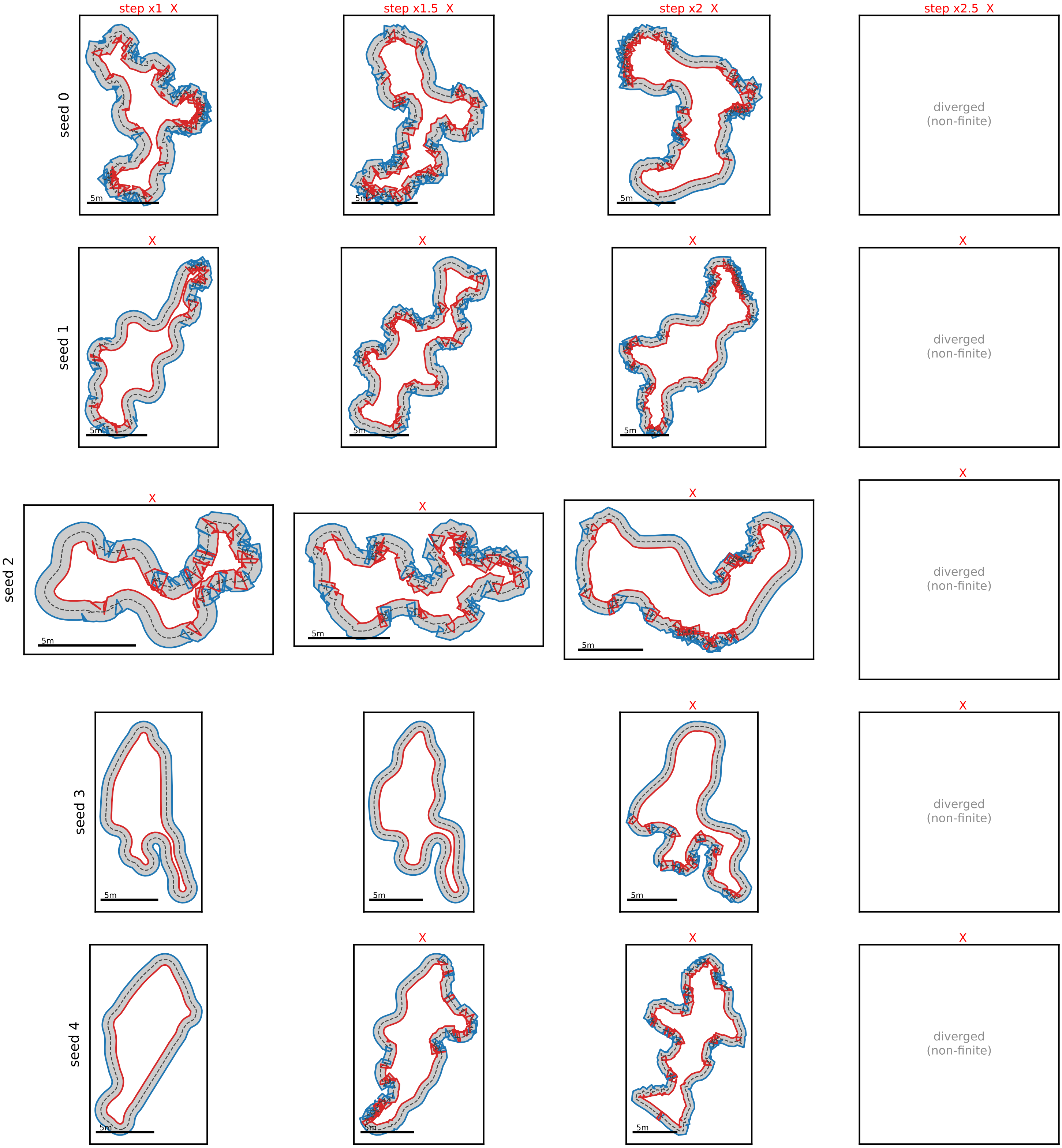

Lever impact on yield and speed (measured E=8192)
Yield vs chain links (iters=150)
(resolution-relative -- see caveats)
valid yield
num_points (chain links)
0.0
0.2
0.4
0.6
0.8
1.0
150
200
250
300
350
400
450
500
Yield vs relax iters (256 links, fair comparison)
valid yield
relax_iters
100
200
300
400
500
600
Yield vs XPBD step scale (256 links, iters=150)
baseline (step x1)
valid yield
sep/spc relaxation factor (over-relaxation)
1.0
1.2
1.4
1.6
1.8
2.0
2.2
2.4
End-to-end speed @ E=8192
relax_iters
s / call (8192 tracks)
num_points
0.5
1.0
1.5
2.0
2.5
3.0
0
100
200
300
400
500
600

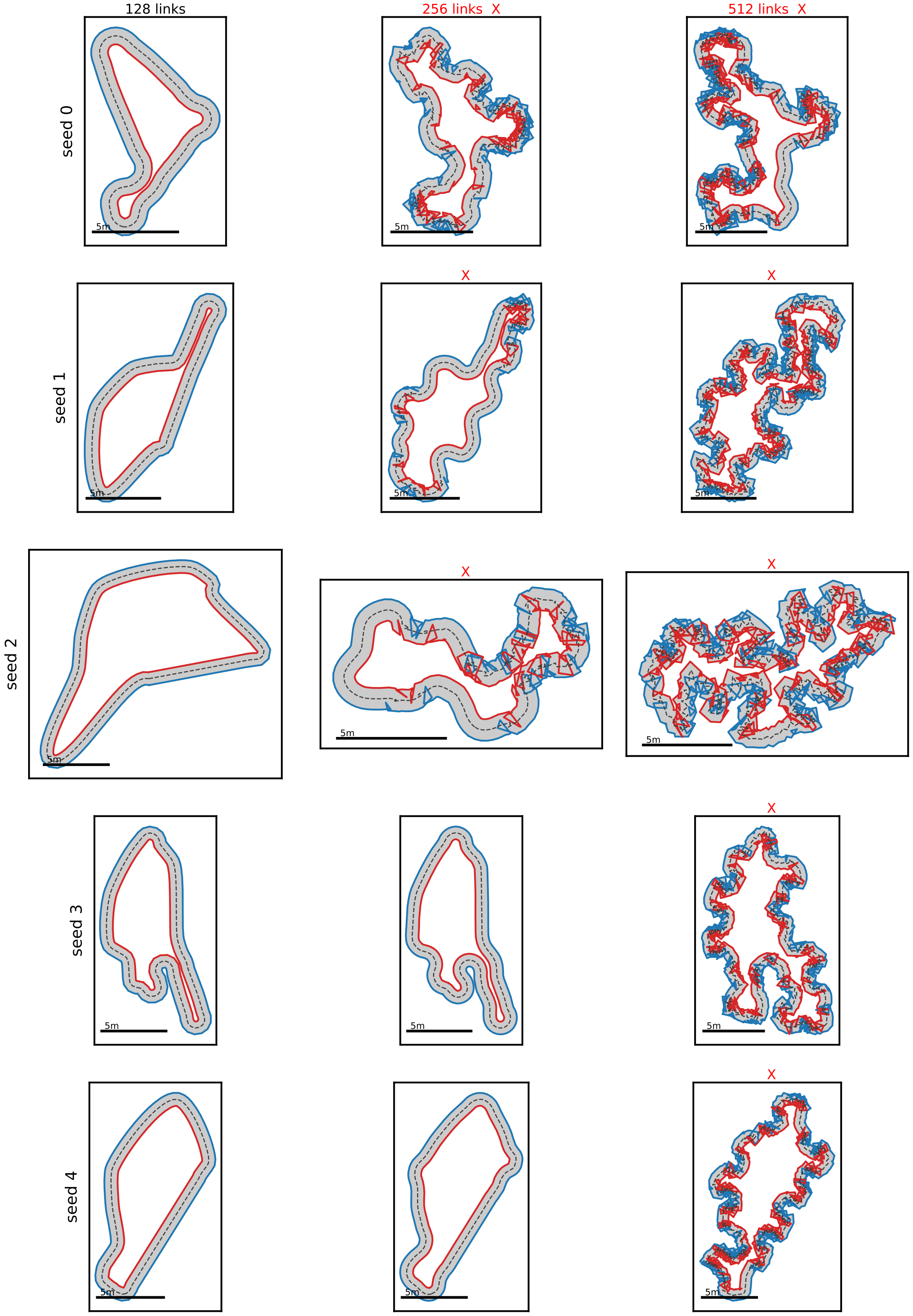
Fixed-seed impact of RELAX ITERS (256 links)
Same generated track per row; columns relax it more. Watch pinched tracks
converge (X = invalid). Yield is convergence-limited.



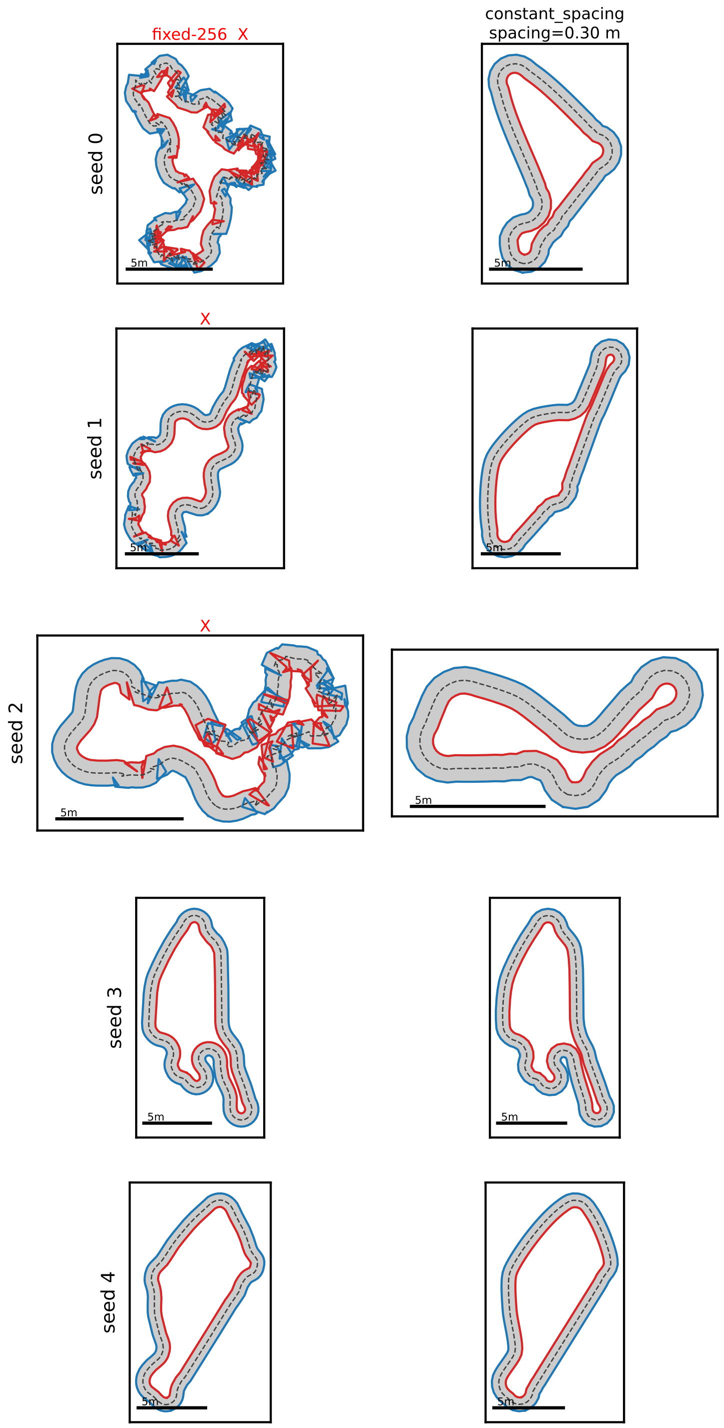
Fixed-seed impact of XPBD STEP SIZE (256 links, iters=150)
Same generated track per row; columns use larger over-relaxation steps (sep/spc factor). The Jacobi solve is unstable for factor > 1: bigger steps OSCILLATE/diverge -> tracks degrade into invalid (X). Larger != faster here.



Fixed-seed impact of CHAIN LINKS (iters=150)
Same seeds; more links = finer relax/validation resolution
(note: the accepted track can differ since the gate
resolution changes).



Constant spacing — the convergence fix
ixed-256 yield=0.684 (E=8192) vs constant_spacing
pacing=0.30 m yield=0.999 (E=8192, 0.557 s/call, 35
IB). Constant spacing is the convergence fix: adaptiv
rc-length resampling to ~0.6×half_width ensures the
PBD relaxation sees a uniform, well-conditioned chai
– eliminating the pinching that fixed-N misses at tigh
bends. X = invalid track.



Findings & recommendation

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FINDINGS (1 m / 20 m regime, E=8192):

* SPEED: baseline 256 links/150 it = 0.79 s/8192 (235 MB). Scales ~O(links^2)
  and ~linear in iters; memory is tiny (<0.5 GB of 16).

* RELAX ITERS -> genuinely raise yield (fair, fixed-256-res): 0.52/0.68/0.77/
  0.83 at 50/150/300/600 it. Yield is RELAXATION-CONVERGENCE-limited.

* XPBD STEP SIZE (over-relaxation) -> larger steps HURT, decisively:
  yield 0.68/0.34/0.02/0.00 at sep/spc factor x1/1.5/2/2.5 (speed unchanged).
  The solve is JACOBI (constraints applied simultaneously), so factor > 1
  oscillates/diverges; x1 is already the stability ceiling. Aiming further
  (margin 0.15->0.30) also drops yield to 0.45. Faster convergence would need
  a better solver (Gauss-Seidel / Chebyshev / chunking), not bigger steps.

* CHAIN LINKS -> more links LOWERS native yield + costs O(N^2). BUT yield is
  RESOLUTION-RELATIVE (the thickness curvature term is sampled at num_points),
  so link counts are NOT directly comparable -- fewer links = looser gate, not
  better tracks. Don't tune links to chase yield.

* REGEN ATTEMPTS -> NO yield change (10/20/40 all 0.684), only slower. The
  regen loop gates on GENERATION simplicity (saturated ~100%); final validity
  is the POST-RELAX check applied after the loop, which regen can't fix.

CAVEATS (adversarially verified):
* Yield is 'valid at the working resolution', not resolution-absolute.
* Re-validating at finer resolution collapses the thickness CURVATURE term
  (artifact of resampling a polyline); SEPARATION stays honest (~0.47 m vs the
  0.49 m target) -- the 1 m-band tracks genuinely do NOT self-overlap.

RECOMMENDATION (1 m/20 m): keep num_points=256; raise relax_iters to ~300-600
for higher honest yield (0.77-0.83). Do NOT enlarge XPBD steps or margin (they
destabilize), and leave max_regen_iters=10 (more is wasted). The ceiling is
relaxation convergence -- the real win is a faster-converging solver.
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